

Successive Differentiation

Define n th derivative of a given variable.

$$y = 3x^4$$

$$\frac{dy}{dx} = 12x^3$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = 36x^2$$

$$\frac{d^3y}{dx^3} = \frac{d}{dx} \left[\frac{d}{dx} \left(\frac{dy}{dx} \right) \right] = 72x$$

The symbols of successive derivative are as follows:-

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \dots, \frac{d^{n-1}y}{dx^{n-1}}, \frac{d^ny}{dx^n}$$

If $y = f(x)$, the successive

derivatives are,

$$f'(x), f''(x), f'''(x), f^{iv}(x), \dots, f^n(x)$$

$$y', y'', y''', y^{iv}, \dots, y^{(n)}$$

$$y_1, y_2, y_3, y_4, \dots, y_n$$

Exercise

$$(2) y = (ax+b)^n$$

$$y' = n(ax+b)^{n-1} \cdot a = an(ax+b)^{n-1}$$

$$y'' = an(n-1)(ax+b)^{n-2} (a) = a^2 n(n-1)(ax+b)^{n-2}$$

$$y''' = a^2 n(n-1)(n-2)(ax+b)^{n-3} (a)$$

$$= a^3 n(n-1)(n-2)(n-3)(ax+b)^{n-3}$$

$$y^{iv} = a^4 n(n-1)(n-2)(n-3)(ax+b)^{n-4}$$

$$\begin{aligned} \therefore y^n &= a^n n(n-1)(n-2)(n-3)\dots(n-(n-1))(ax+b)^{n-n} \\ &= a^n n(n-1)(n-2)\dots 1(ax+b)^0 \\ &= a^n n! \end{aligned}$$

$$(3) y = \ln(ax+b)$$

$$y' = \frac{a}{ax+b} = a(ax+b)^{-1} = a^1 0! (ax+b)^{-1}$$

$$y'' = a(-1)(ax+b)^{-2} \cdot a = -a^2(ax+b)^{-2} = -1! a^2(ax+b)^{-2}$$

$$y''' = -a^2(-2)(ax+b)^{-3} \cdot a = 2a^3(ax+b)^{-3} = 2! a^3(ax+b)^{-3}$$

$$\begin{aligned} y^{iv} &= \cancel{2a^3}(ax+b)^{-4}(-3)(a) \\ &= -6a^4(ax+b)^{-4} = -3! a^4(ax+b)^{-4} \end{aligned}$$

$$\begin{aligned} y^{5v} &= -6a^4(-4)(ax+b)^{-5} = 24a^5(ax+b)^{-5} \\ &= 4! a^5(ax+b)^{-5} \end{aligned}$$

$$\therefore y^n = (-1)^{n-1} (n-1)! a^n (ax+b)^{-n}$$